

Bruggemann (2021) - Higher Taxes at the Top: The Role of Entrepreneurs

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September 30, 2026

1 Model description

This is a brief description of the replication of Brüggemann (2021), hereafter B2021. We make very few notational changes. It is not intended to be a full description of the model; rather, the aim is to make clear how the model is expressed in terms of the VFI Toolkit. For a full description of the model, see the original paper. The overall takeaway is that everything replicates.

The model is a general equilibrium incomplete-markets infinite-horizon model. The value function problem has two decision variables (labor supply and entrepreneurship), one endogenous state (assets), and three exogenous states (age, labor productivity, and entrepreneurial productivity).¹ Agents decide each period whether to be entrepreneurs or workers.

Households can be young or old, and age stochastically. While young they can be workers or entrepreneurs; when old they are retirees. Agents have labor productivity (per unit of time worked), η , and entrepreneurial productivity, θ .²

When young, households decide whether to be workers or entrepreneurs. Workers choose labor supply and savings. Entrepreneurs choose savings, while their capital demand, hired labor, and output are determined by a static production problem conditional on the household state and prices.³ Because this static problem has an analytical solution, capital and hired labor do not appear explicitly as value function decision variables below.

Let $e = 0$ denote worker and $e = 1$ denote entrepreneur. Let $age = 1$ denote young and $age = 2$ denote old.

The construction of the exogenous-state grid and transition matrix is described in Section 1.1, and the entrepreneur's static problem is described in Section 1.3.

¹You do not actually need to include entrepreneurship as a decision variable, and doing so here slows the code down. But without it you are constantly checking whether the agent chooses worker or entrepreneur, making the code messier and more prone to typographical errors. Since the model solves fine either way, it is much easier to treat worker-entrepreneur status as a binary-valued decision variable.

²Because η and θ are only needed for the young, we store age, labor productivity, and entrepreneurial productivity on a single exogenous-state grid whose rows enumerate the relevant combinations of these variables. Section 1.1 explains this construction in detail.

³Entrepreneurs are forced to supply a certain amount of their own labor, and get disutility from this.

The household value function problem is given by

$$V(a, age, \eta, \theta) = \max_{a', l, e} \left\{ \frac{c^{1-\sigma_1}}{1-\sigma_1} - \xi \frac{l^{1+\sigma_2}}{1+\sigma_2} + \beta E[V(a', age', \eta', \theta') \mid age, \eta, \theta] \right\}$$

subject to

Young ($age = 1$): choose $e = 0$ (worker) or $e = 1$ (entrepreneur),
whichever gives higher utility,

Worker ($e = 0$): $I = wl\eta + ra$

$$(1 + \tau_c)c + a' = I + a - TaxI(I)$$

$$a' \geq 0$$

Entrepreneur ($e = 1$): $I = y - \delta k - r(k - a) - wn$

$$(1 + \tau_c)c + a' = I + a - TaxI(I)$$

$$y = \theta (k^\gamma (\bar{l} + n)^{1-\gamma})^v$$

$$k \leq \lambda a$$

$$l = \bar{l}$$

$$a' \geq 0, \quad n \geq 0$$

Old ($age = 2$): $I = pension + ra$

$$(1 + \tau_c)c + a' = I + a - TaxI(I)$$

$$a' \geq 0, \quad l = 0$$

where $TaxI(I)$ is a tax function on income I .

Note that workers can choose labor supply l , while entrepreneurs must supply $\bar{l}$. Entrepreneurs' η is not relevant for the labor they supply to their own businesses, but they still receive the disutility from supplying $l = \bar{l}$.⁴ The interpretation is that entrepreneurs always supply own labor $\bar{l}$, while hired labor n is chosen subject to $n \geq 0$. Section 1.3 clarifies how this restriction enters the static entrepreneur problem.

The above household problem is deliberately compressed because it omits y , k , and n from the list of choices to be made. Since the entrepreneur's production problem is static and has an analytical solution, these variables are solved for directly, done in the codes as `B2021_EntrepreneurStaticProblem()`, rather than being included as additional value function choices. They are nevertheless shown above in the equations (just not as decisions/choices) to make the entrepreneur's budget constraint and feasibility restrictions explicit.

The analytical solution to the entrepreneurial production decision follows. The optimal output (y), capital demand (k), and labor demand (n) can be written as functions of the rental rate of capital r , the wage w , the capital depreciation rate δ , the collateral constraint parameter λ , and the production function parameters θ , γ , and v . See Section 1.3 for a full derivation and the discussion at discourse.vfitoolkit.com/t/on-solving-models-with-entrepreneurs/296

The main exercises in Bruggemann (2021) are to solve for the initial stationary equilibrium under the existing US tax system. We then vary the statutory top marginal tax rate, $\tau_{i,r6,stat}$, and

⁴By giving entrepreneurs the disutility of labor supply, and by calibrating $\bar{l}$ to equal the average l of workers, B2021 makes endogenous labor largely irrelevant for the decision to be a worker or entrepreneur. This still does not explain why entrepreneur η is irrelevant to the entrepreneur's own labor supply, which appears to be imposed exogenously. In Kitao (2008), by contrast, entrepreneur η enters the entrepreneur's own labor supply, although that model has exogenous labor supply.

for each value solve for the final stationary equilibrium and the transition path. The statutory rate runs from 35% (the benchmark) to 100% in steps of five percentage points. Since the effective rate is $\tau_{i,adj}$ times the statutory one with $\tau_{i,adj} = 0.669$, this covers effective top marginal rates from 23.4% to 66.9%, essentially the same range as the 23.5% to 70% of the original's Figure 1. For the alternative economies of Figure 6 the statutory rate is extended to 120%, i.e. effective rates up to 80.3%, so that those welfare curves have an interior peak on the same axis the original uses. For the rate that maximizes welfare we create more detailed model outputs.

For the stationary general equilibrium, aggregate assets finance entrepreneurial capital and corporate capital, and worker effective labor is split between entrepreneurial hired labor and corporate labor. Thus the code constructs $K_{corp} = A - K_{noncorp}$ and $N_{corp} = L - N_{\bar{l}} - N_{noncorp}$, where $N_{\bar{l}}$ is the entrepreneurs' own labor. The own-labor term has to be subtracted because L is the total supply of efficiency units of labor, including the $\bar{l}$ that each entrepreneur supplies to their own business, and that labor is neither hired on the market nor available to the corporate sector; it is already an input inside $Y_{noncorp}$. The original does the same thing, building corporate labor as `totlcorp=toteffl-hiredlabe` (`benchmark.f90`). The equations for capital market clearance, labor market clearance, and government budget are thus respectively,

$$r = \left[\alpha Z \left(\frac{K_{corp}}{N_{corp}} \right)^{\alpha-1} - \delta \right]$$

$$w = (1 - \alpha) Z \left(\frac{K_{corp}}{N_{corp}} \right)^{\alpha}$$

$$G + PensionSpending + Lumpsum = IncomeTaxRevenue + ConsumptionTaxRevenue$$

The corresponding general-equilibrium variables are the interest rate r , the wage w , and the linear income tax rate τ_s , where τ_s adjusts to balance the government budget.⁵ The same general-equilibrium command can also be used for calibration targets, here calibrating G , $pension$ and $\bar{y}$ through a government-spending share, a pension replacement rate, and an average-income condition. Hence the initial solve uses six conditions in total: three market-clearing or budget-balance conditions that must hold in any competitive equilibrium, plus three calibration conditions, solved jointly for $\{r, w, \tau_s, G, pension, \bar{y}\}$. Note that G and $pension$ are calibrated as *ratios* (to output and to average income respectively), which is what the original does: its `benchmark.f90` sets government spending as `gfrac*gdp` with `gfrac=0.146` and the retirement benefit as `replrate*totinc` with `replrate=0.4`. Having calibrated those three, a second solve holds them fixed and finds $\{r, w, \tau_s\}$ alone, so that the reported benchmark is a competitive equilibrium in the ordinary sense. In the counterfactual equilibria τ_s is instead held at its benchmark value and a lump-sum transfer absorbs the extra revenue, so the general-equilibrium variables there are $\{r, w, lumpsum\}$.

Notational changes: B2021 called worker productivity ϵ which we denote here as η .

For more details on the model see Brüggemann (2021). The rest of this doc was largely written by Claude with light human editing.

⁵Because the corporate Cobb-Douglas technology has constant returns to scale, stationary equilibrium does not require treating r and w separately: given r , the capital-labor ratio K_{corp}/N_{corp} follows from the marginal product of capital, and then w follows from the marginal product of labor.

1.1 Grid and transition matrix for exogenous state z

The exogenous state is

$$z = [age, \eta, \theta],$$

where $age \in \{1, 2\}$, with $age = 1$ denoting “young” and $age = 2$ denoting “retired”, $\eta \in \{\bar{\eta}_1, \dots, \bar{\eta}_{n_\eta}\}$ is labor productivity, and $\theta \in \{\bar{\theta}_1, \dots, \bar{\theta}_{n_\theta}\}$ is entrepreneurial productivity. If we used a standard cross-product grid it would be wasteful, as there is no point tracking η and θ for the old. We therefore use a joint grid, which we now describe. In the code, z (denoted as z_grid) is an array of size $[n_\eta \times n_\theta + 1, 3]$. The first $n_\eta \times n_\theta$ rows correspond to young agents, with η varying fastest, and the last row corresponds to the old state. Since retired agents do not use (η, θ) , the code stores the old state as $(2, 0, 0)$. Thus the total number of states is

$$N_z \equiv n_\eta \times n_\theta + 1.$$

Assuming for simplicity that $n_\theta = 2$, the array z_grid is

$$z_grid = \begin{bmatrix} 1 & \bar{\eta}_1 & \bar{\theta}_1 \\ 1 & \bar{\eta}_2 & \bar{\theta}_1 \\ \vdots & \vdots & \vdots \\ 1 & \bar{\eta}_{n_\eta} & \bar{\theta}_1 \\ 1 & \bar{\eta}_1 & \bar{\theta}_2 \\ 1 & \bar{\eta}_2 & \bar{\theta}_2 \\ \vdots & \vdots & \vdots \\ 1 & \bar{\eta}_{n_\eta} & \bar{\theta}_2 \\ 2 & 0 & 0 \end{bmatrix}$$

For arbitrary n_θ ,

$$\underbrace{z_grid}_{[n_\eta \times n_\theta + 1, 3]} = \begin{bmatrix} \mathbf{1}_{N_y} & \mathbf{1}_{n_\theta} \otimes \eta & \theta \otimes \mathbf{1}_{n_\eta} \\ 2 & 0 & 0 \end{bmatrix}$$

where $\mathbf{1}_m$ denotes an $m \times 1$ vector of ones and $N_y \equiv n_\eta \times n_\theta$ is the number of young states. The values (η, θ) in the last row are placeholders only.

The joint transition matrix for z is built from the separate Markov chains for labor and entrepreneurial productivity, defined as $\pi_\eta(\eta, \eta')$ and $\pi_\theta(\theta, \theta')$, together with the Markov chain for age . Using the notation in the MATLAB code,

$$\begin{bmatrix} t \setminus t+1 & age' = 1 \text{ (young)} & age' = 2 \text{ (retired)} \\ age = 1 \text{ (young)} & \pi_y & 1 - \pi_y \\ age = 2 \text{ (retired)} & 1 - \pi_o & \pi_o \end{bmatrix}$$

where $1 - \pi_y$ is the probability that a young agent becomes retired and $1 - \pi_o$ is the probability that an old agent dies and is replaced by a young agent.

The η grid is chosen by taking the first five values of the labor-productivity process and their transition matrix from (Cagetti and De Nardi, 2009), and then adding a sixth “superstar” state. Thus

$$\text{eta_grid} = [0.2468 \quad 0.4473 \quad 0.7654 \quad 1.3097 \quad 2.3742 \quad 26]^T,$$

and π_η is the corresponding 6×6 transition matrix constructed in the code from the Cagetti–De Nardi matrix plus the additional superstar transition probabilities.

The entrepreneurial-ability grid is also hard-coded, using the values reported in B2021 (Table 2):

$$\text{theta_grid} = [0 \quad 0.682 \quad 1.750 \quad 2.818]^T,$$

with π_θ given by the 4×4 matrix in the MATLAB code.

Now that we have $\pi_\eta(\eta, \eta')$, $\pi_\theta(\theta, \theta')$, and the unconditional distributions of η and θ , denoted by π_η^* and π_θ^* , we can build the transition matrix for $z = [age, \eta, \theta]$. First, combine $\pi_\eta(\eta, \eta')$ and $\pi_\theta(\theta, \theta')$ with a Kronecker product. Since η varies fastest, the order is

$$\Pi_{young}((\eta, \theta), (\eta', \theta')) = \pi_\theta(\theta, \theta') \otimes \pi_\eta(\eta, \eta').$$

Then, letting $N_y \equiv n_\eta \times n_\theta$ denote the number of young states, the joint transition matrix is

$$\pi_z(z, z') = \begin{bmatrix} \pi_y \Pi_{young} & (1 - \pi_y) \mathbf{1}_{N_y} \\ (1 - \pi_o) \Pi_{newborn} & \pi_o \end{bmatrix}$$

where the $1 \times N_y$ row vector $\Pi_{newborn}$ is

$$\Pi_{newborn} = \pi_\theta^*(\theta') \otimes \pi_\eta^*(\eta').$$

Newborn young agents draw (η, θ) independently from their stationary distributions, and old agents either remain old or are replaced by such newborn agents.

1.2 Decision variables

There are two decision variables, labor supply l and worker/entrepreneur e . Because only workers choose labor supply we can again use a joint-grid on decision variables (all the labor choices with $e = 0$, and then just a fixed labor supply with $e = 1$). This reduces the number of grid points from $2n.l$ to $n.l + 1$.

1.3 Entrepreneurs static problem

This part is implemented in the MATLAB function *B2021_EntrepreneurStaticProblem*. Entrepreneurs solve the following static maximization problem:

$$\pi(a, \theta) = \max_{k, n} \left\{ \theta \left(k^\gamma (\bar{l} + n)^{1-\gamma} \right)^v - (r + \delta)k - wn \right\}$$

s.t.

$$\begin{aligned} k &\leq \lambda a, \\ n &\geq 0, \end{aligned}$$

where $\bar{l}$ is a fixed parameter. For convenience, define total labor input by $\bar{n} = \bar{l} + n$, so the constraint $n \geq 0$ is equivalent to $\bar{n} \geq \bar{l}$. We distinguish two cases according to whether the nonnegativity constraint on hired labor binds.

1. Assume first that the hired-labor constraint does not bind, so $n > 0$ (equivalently, $\bar{n} > \bar{l}$). Then the unconstrained optimal capital is

$$k^{unc} = (\theta v)^{\frac{1}{1-v}} \left(\frac{\gamma}{r + \delta} \right)^{1 + \frac{\gamma v}{1-v}} \left(\frac{1 - \gamma}{w} \right)^{\frac{v(1-\gamma)}{1-v}}$$

Taking the collateral constraint into account gives

$$k^* = \min \{k^{unc}, \lambda a\}.$$

The optimal $\bar{n} = \bar{l} + n$ is given by

$$\bar{n}(k^*) = \left(\frac{\theta(1-\gamma)v}{w} \right)^{\frac{1}{1-(1-\gamma)v}} (k^*)^{\frac{\gamma v}{1-(1-\gamma)v}}.$$

Finally, we must check whether $\bar{n} > \bar{l}$ (equivalently, $n > 0$). If this is the case, then the solution is

$$\begin{aligned} k(a, \theta) &= k^*, \\ n(a, \theta) &= \bar{n} - \bar{l}. \end{aligned}$$

Compute $\pi(a, \theta)$ by replacing k and n with their optimal values in

$$\pi(a, \theta) = \theta \left(k^\gamma (\bar{l} + n)^{1-\gamma} \right)^v - (r + \delta)k - wn.$$

If instead it turns out that $\bar{n} \leq \bar{l}$, then the candidate interior solution violates the constraint $n \geq 0$, so we go to the second case.

2. Suppose now that the hired-labor constraint binds. Then $n = 0$ and $\bar{n} = \bar{l}$, so the problem becomes

$$\pi(a, \theta) = \max_{k \leq \lambda a} \left\{ \theta \left(k^\gamma (\bar{l})^{1-\gamma} \right)^v - (r + \delta)k \right\}$$

The unconstrained level of capital is

$$k^{unc} = (\theta v)^{\frac{1}{1-\gamma v}} \left(\frac{\gamma}{r + \delta} \right)^{\frac{1}{1-\gamma v}} (\bar{l})^{\frac{v(1-\gamma)}{1-\gamma v}}.$$

Taking the collateral constraint into account gives

$$k^* = \min \{k^{unc}, \lambda a\}.$$

The solution is then

$$\begin{aligned} k(a, \theta) &= k^*, \\ n(a, \theta) &= 0. \end{aligned}$$

Compute $\pi(a, \theta)$ by replacing k and n with their optimal values in

$$\pi(a, \theta) = \theta \left(k^\gamma (\bar{l} + n)^{1-\gamma} \right)^v - (r + \delta)k - wn.$$

In Brüggemann (2021), the same problem is solved by taking taxes explicitly into account:

$$\max_{k,n} y^e - T(y^e)$$

s.t.

$$\begin{aligned} k &\leq \lambda a, \\ n &\geq 0, \\ y^e &= \theta \left(k^\gamma (\bar{l} + n)^{1-\gamma} \right)^v - (r + \delta)k - wn + ra. \end{aligned}$$

where y^e denotes entrepreneurial taxable income and $T(\cdot)$ is the income tax function. She computes optimal labor demand as a function of capital as follows:

$$\bar{n}(k) = \left(\frac{\theta (1 - \gamma) v}{w} \right)^{\frac{1}{1 - (1 - \gamma)v}} k^{\frac{\gamma v}{1 - (1 - \gamma)v}},$$

so that

$$n = \max \{ \bar{n} - \bar{l}, 0 \}.$$

She then optimizes with respect to capital k using grid search. In particular, n , y^e , and $y^e - T(y^e)$ are computed for each grid value of k , and she picks the maximum of the array $y^e - T(y^e)$.

2 Comments on Replication results

The replication is successful. Both the initial stationary general equilibrium and the core policy result come out close to the original.

The central number in B2021 is the welfare-maximizing effective top marginal tax rate (TMTR). She reports 60 percent when welfare is evaluated over the whole transition path, and 57 percent when only the two steady states are compared. Our replication gives 60.2 percent and 56.9 percent respectively. The statutory rates behind these are 90 and 85 percent; the gap between statutory and effective rates is the adjustment factor $\tau_{i,adj} = 0.669$ that B2021 uses to reconcile statutory tax law with effective rates actually paid, and it is important to keep track of which of the two any given number refers to. The peak CEV is 8.7 percent on the transition criterion and 5.4 percent on the steady-state criterion. The appendix Laffer curves agree as well: revenue is maximized at an effective 63.6 percent in the steady state against her 62 percent, and at 60.2 percent in the first period of the transition against her 60 percent.

Figure 1 is the core result. The replication reproduces the shape of the original: welfare gains rise steeply at first, flatten out between roughly 54 and 64 percent, and turn down thereafter, so the maximum is interior on both criteria. On the transition criterion the CEV is within 0.15 percentage points of its peak everywhere between 56.9 and 63.6 percent, which is why the exact location of the maximum is not sharply identified by a grid of five-percentage-point steps in the statutory rate; the same caveat applies to the original. Figure 2 shows the same two curves against the statutory rather than the effective tax rate. Figure 9 does the same for the Laffer curve of the original's appendix.

Tables 11 to 14 concern the initial stationary general equilibrium, and the replication is close throughout. One point is worth stressing when reading them: the “Target” column is data, and the original misses several of those targets itself, so the honest comparison for a replication is against the “Model” column of the original paper rather than against the target. On that comparison the entrepreneurs’ share of total income (0.22 against her 0.22, both against a target of 0.17) and the entrepreneurs’ income Gini (0.62 against her 0.62, target 0.65) are exact, and the whole of Table 14 reproduces her model column, which likewise sits some distance from the SCF data. The capital-output ratio is 2.67 against her 2.65, the top 1 percent income share, the wealth Gini, the fraction of entrepreneurs, the entry rate and the share of entrepreneurs among the top 1 percent of income all match, and the share of entrepreneurs who hire matches to three decimals. The exit rate is 0.24 against her 0.22; hers is computed exactly from the occupation state, whereas ours has to be estimated from a simulated panel, so some difference is expected.

Two rows do not match, and in both cases we know why.

The first is the average federal income tax rate in Table 11, and with it the average tax rates for top percentiles in Table 13, which come out roughly four percentage points above the original. This is a deliberate difference of definition: we count the revenue from the linear income tax τ_s as federal income tax, since it is a federal tax levied on income, whereas the original does not. Her `benchmark.f90` computes the average tax rate as `DOT.PRODUCT(dist,inctax)/totinc` and her `table5.do` sums `inctax*dist`, and in both cases `inctax` is the bracketed income tax alone; the τ_s revenue sits in a separate vector that neither statistic touches.

The second is the ratio of median net worth of entrepreneurs to that of everyone else, which we get as 5.7 against her 7.7 (the target is 7.26, so both fall short, but hers by much less). The cause is a genuine difference between the two implementations, and the note to Table 11 sets it out. In this replication occupation is a decision, so our entrepreneurs are the young households who choose to be entrepreneurs this period. In the original occupation is part of the state, and the group she takes the median over is the mass sitting in the young-entrepreneur block of the state space, i.e. the households who chose to be entrepreneurs *last* period. Her group therefore leaves out this period’s entrants and keeps this period’s exiters, while ours does the opposite. Entrants are around a fifth of the entrepreneur mass and are drawn from the worker asset distribution, so excluding them raises the median considerably. Matching her number would require putting occupation back into the state.

Tables 15 to 18 and Figures 4 to 7 are based on the transition to the welfare-maximizing TMTR, which for us is an effective 60.2 percent against her 60 percent, so the two experiments are almost the same experiment and the tables can be compared directly. Table 15 replicates well in general equilibrium: output, capital, labor, the interest rate and the wage all come within a few tenths of a percentage point of the original. The weakest entry is tax revenue, which rises by 17.0 percent against her 19.6 percent in general equilibrium and 13.4 against 15.5 in partial equilibrium; this is presumably related to the treatment of τ_s noted above. Table 16 reproduces every one of its eight entries to within 0.8 percentage points, with each of them erring in the direction the slightly larger tax increase would imply. Table 17 reproduces to within 0.2 percentage points in every cell.

Table 17 needs one remark about the labor columns. They are shares of *market* labor, so entrepreneurs’ own inelastically supplied $\bar{l}$ hours are excluded from the denominator, exactly as in the construction of corporate labor described in the model section. Dividing by total hours instead gives 56.9 and 43.1 against her 60.5 and 39.5, whereas dividing by market labor gives 60.7 and 39.3, so this is clearly the convention the original uses.

Figure 4 reports welfare results for “workers” and “entrepreneurs”. The economy also includes a third group omitted from the figures, namely the old. In the replication we decided to include the old to give a fuller picture, but we had to omit them from the second panel because the welfare gains of the old in the “Top 3%” were so large that they ruin the graph, being more than 100 times as large as any gains for the worker and entrepreneur groups.

Figure 8 shows the welfare-maximizing TMTR in the “no-entrepreneur” models, and Figure 10, which is not in the original, shows the corresponding Laffer curves, so that Figure 10 can be compared with Figure 9 just as Figure 8 can be compared with Figure 1. The qualitative result of the original is reproduced: removing entrepreneurs raises the welfare gains from taxing the top substantially, from a peak CEV of 5.4 percent to 7.9 percent on the steady-state criterion, and restricting the highest entrepreneurial ability lowers them to 2.3 percent. The one place where we do not reproduce the original is the *location* of the no-entrepreneur peak: she finds the welfare-maximizing effective rate rising to approximately 67 percent once entrepreneurs are removed, whereas ours stays at 60.2 percent, the same rate as the benchmark economy. We extended the tax grid to an effective 80.3 percent precisely so that this could not be an artifact of the peak sitting at the edge of the plotted range, and it is not. So we reproduce her conclusion that entrepreneurs dampen the welfare gains from higher top taxes, but not her further conclusion that their presence pushes the optimal rate itself down.

3 Numerical implementation

B2021 uses pure discretization, inherited from the Cagetti and De Nardi (2009) codes she builds on, to solve the value function problem. Her Fortran codes put 300 points on endogenous labor supply, which limits how many can be spent on assets, and she ends up with 480 points on assets (`benchmark.f90` lines 1102–4). Our replication uses 300 points on labor supply purely to match her, and 501 points on assets, plus a grid interpolation layer that interpolates next-period assets at 50 equally spaced points between every two consecutive asset grid points. Realistically, for a model like this, 100 points on labor supply would be enough for almost everything, and 300 is mostly wasting compute and memory that would be better spent on assets.

The transition paths are solved with Anderson acceleration. Two details of the original’s statistics are worth recording because they are easy to get wrong when comparing against it, and both were only found by reading her code rather than the paper. First, most of the statistics in Tables 11, 13 and 14 and in Figures 5 and 7 are not computed in the Fortran at all: it writes Stata data files and the definitions live in `Stata/Dofiles`. Only the capital-output ratio, average hours, the median net worth ratio, the fraction of entrepreneurs, the entrepreneurs’ income and asset shares, the entry and exit rates, the fraction hiring and the average tax rate come out of the Fortran itself. Second, her do-files classify a household as an entrepreneur by `kstar>0`, which puts the retirees in with the workers; we follow that classification, which is why the workers’ income Gini here is taken over young workers and retirees together rather than over young workers alone.

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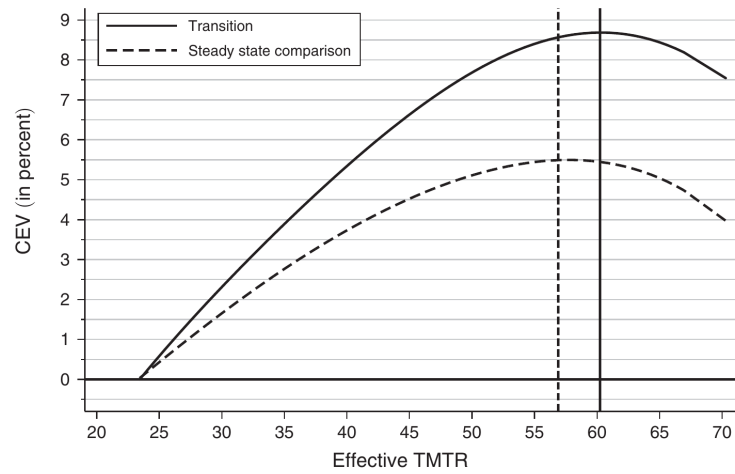


FIGURE 1. WELFARE-MAXIMIZING TOP MARGINAL TAX RATE

Notes: The solid line shows the CEVs (in percent) associated with increasing the top marginal tax rate to the levels indicated on the x-axis, taking into account the whole transition path. The dashed line shows the CEVs (in percent) when comparing steady states.

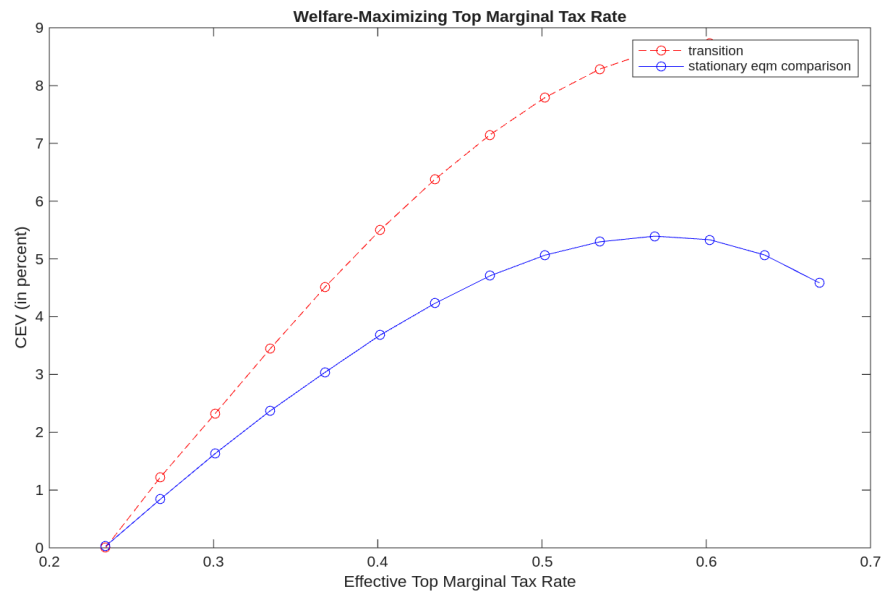


Figure 1: Figure 1 of Bruggemann (2021)

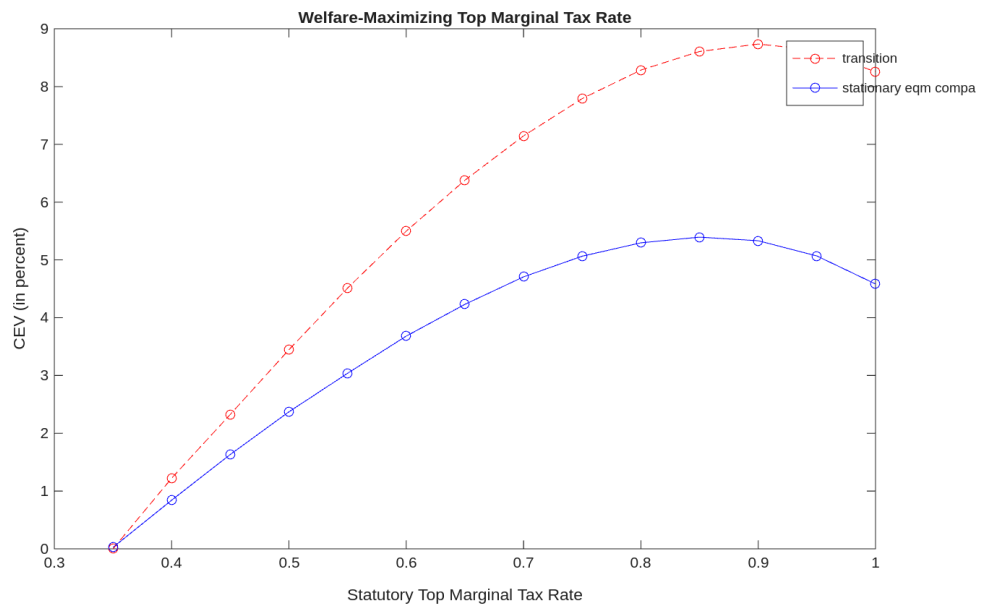


Figure 2: Figure 1 of Bruggemann (2021) redrawn against the *statutory* top marginal tax rate rather than the effective one. The two curves are the same as in Figure 1; only the x-axis differs, by the factor $\tau_{i,adj} = 0.669$.

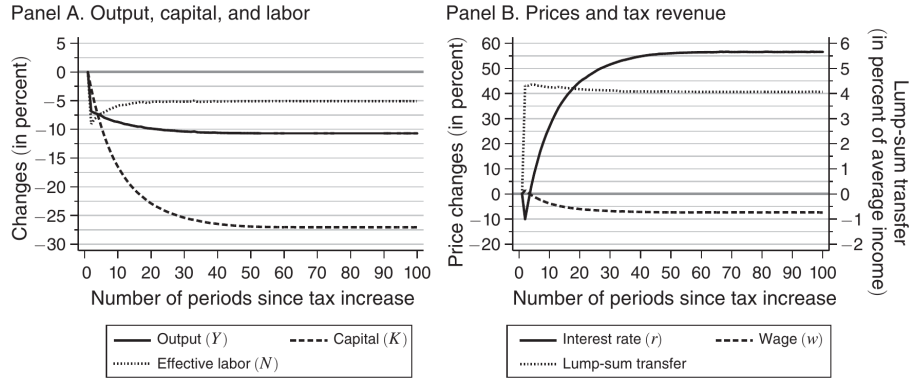


FIGURE 2. CHANGES IN AGGREGATE VARIABLES ALONG THE TRANSITION

Notes: The solid, dashed, and dotted lines in panel A show percentage changes in output, capital, and effective labor, respectively, along the transition between the initial steady state (period 0) and the higher-tax equilibrium. The solid and dashed lines in panel B show percentage changes in interest rate and wage, respectively, along the transition. The dotted line in panel B shows the lump-sum transfer paid out along the transition in percent of average income.

Changes in Aggregate Variables along the Transition

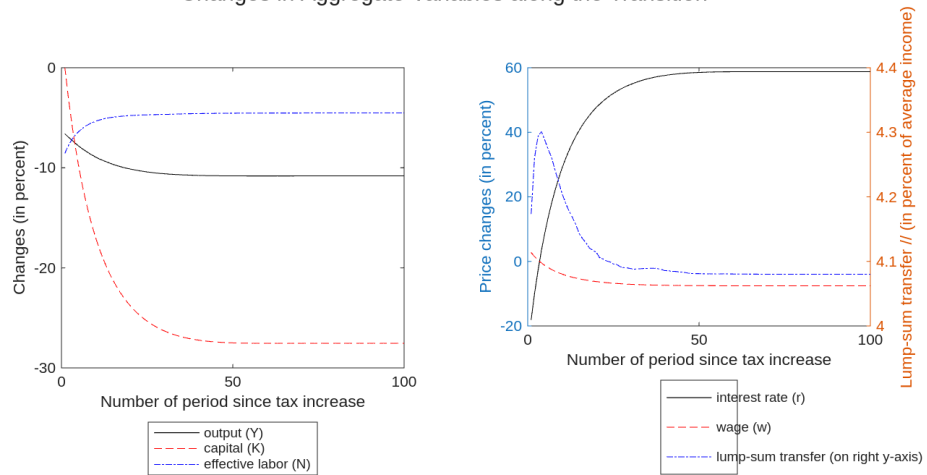


Figure 3: Figure 2 of Bruggemann (2021)

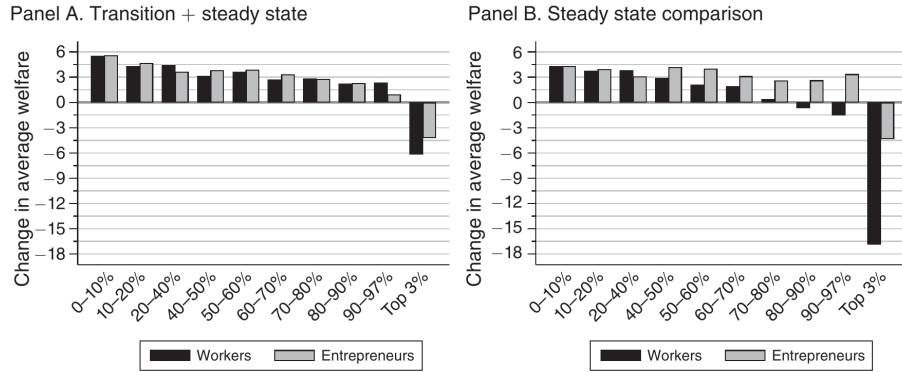


FIGURE 3. WELFARE GAINS BY INCOME DECILES FOR WORKERS AND ENTREPRENEURS

Notes: The black bars show the average changes in welfare for workers as measured by changes in households' expected lifetime utility for each decile of the distribution of gross income. The gray bars show the same for entrepreneurs. Panel A shows the welfare changes when taking into account the transition path to the new steady state. Panel B compares the two steady states only. A household is characterized by its occupation, age, and endowments with wealth, labor, and entrepreneurial ability.

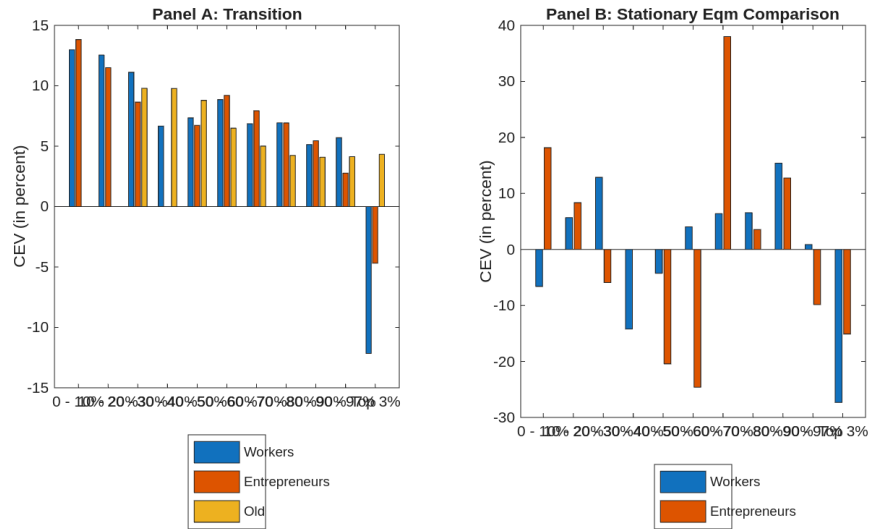


Figure 4: Figure 3 of Bruggemann (2021)

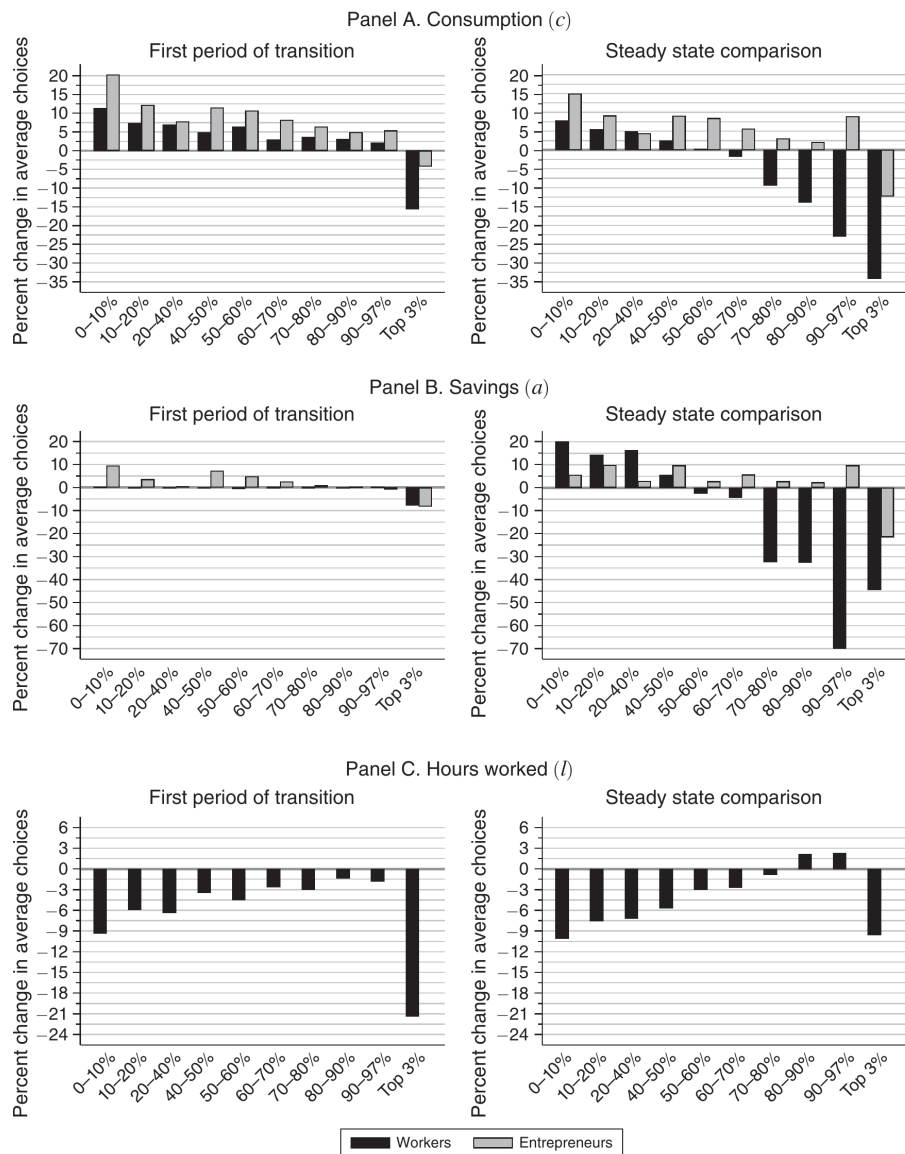
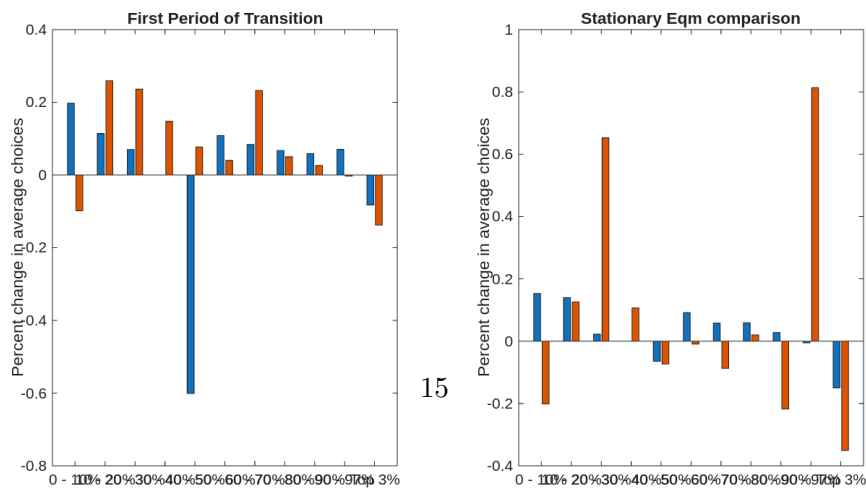


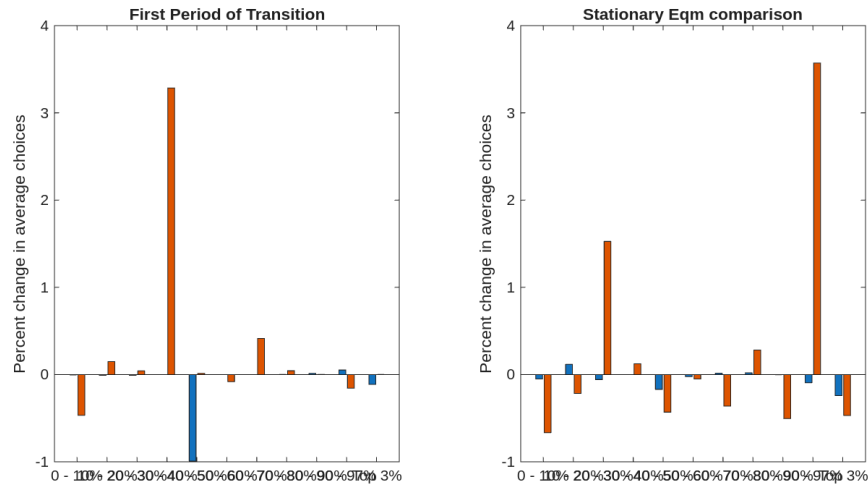
FIGURE 4. CHANGES IN AVERAGE CHOICES BY OCCUPATION AND INCOME LEVEL

Notes: The black bars show the average changes in workers' consumption (panel A), savings (panel B), and hours worked (panel C) for each decile of the distribution of gross income after the TMTR was increased to 60 percent. The gray bars show the same for entrepreneurs. The left panels in each subfigure depict changes in the first period of the transition; the right panels show changes between steady states.

Consumption (c)



Savings (a)



Hours Worked (l)

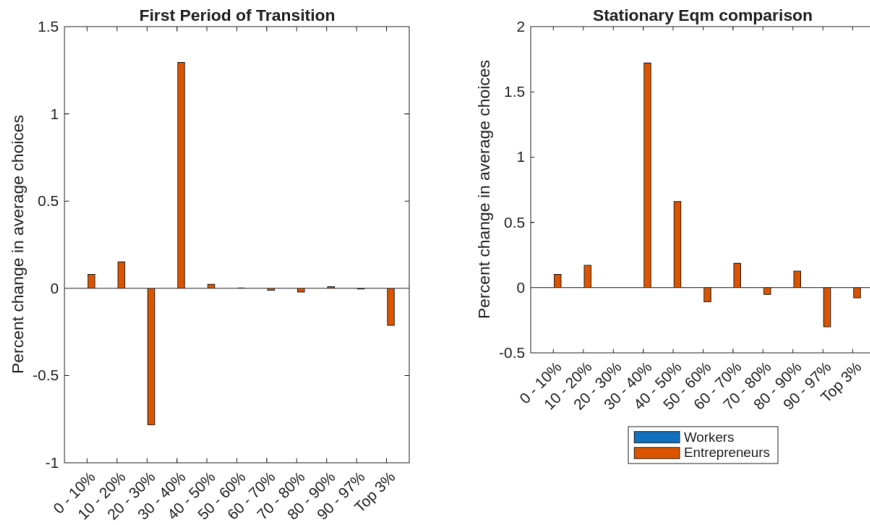


Figure 6: Figure 4 of Bruggemann (2021): part 2/2

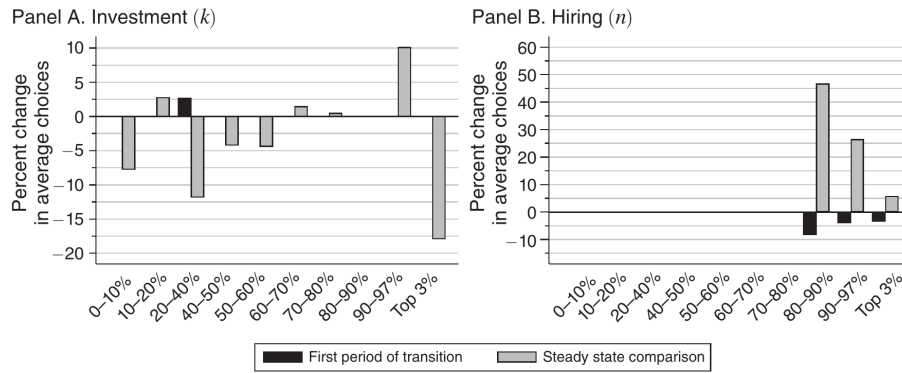


FIGURE 5. CHANGES TO ENTREPRENEURS' AVERAGE CHOICES BY INCOME LEVEL

Notes: Panel A shows percentage changes in entrepreneurs' investment for each decile of the distribution of gross income after the TMTR was increased to 60 percent; panel B shows changes in hiring. The black bars depict changes in the first period of the transition; the gray bars show changes between steady states.

Changes to Entrepreneurs Average Choices by Income Level

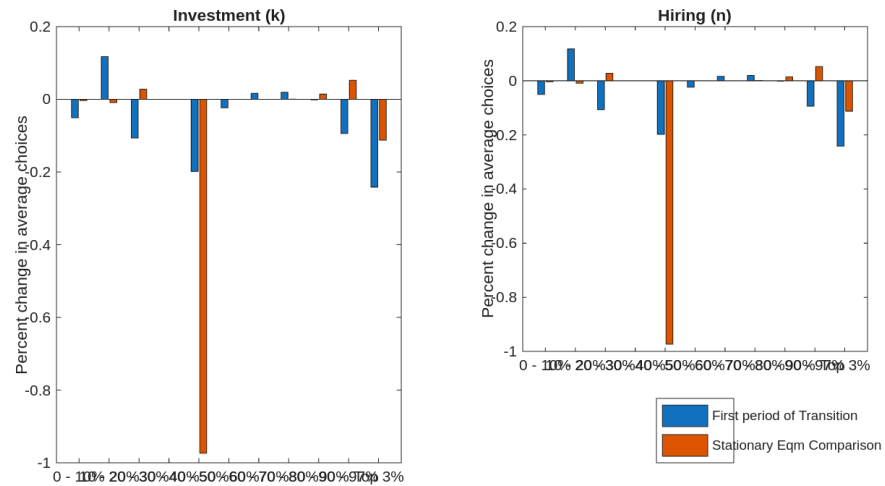


Figure 7: Figure 5 of Bruggemann (2021)

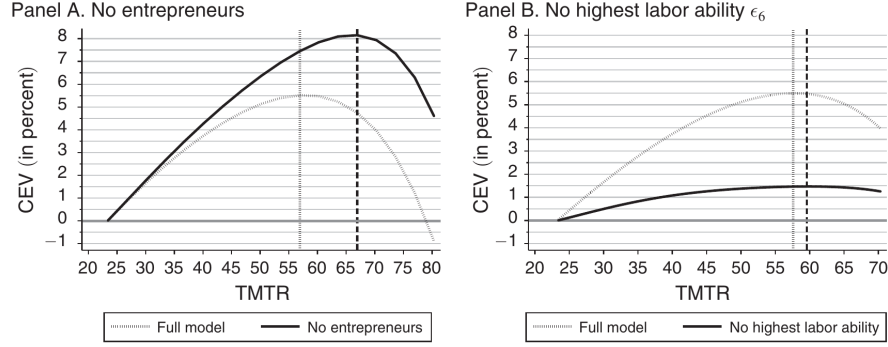


FIGURE 6. OPTIMAL TAX RATES UNDER ALTERNATIVE SPECIFICATIONS

Notes: In panel A, the solid line shows the CEVs (in percent) associated with increasing the top marginal tax rate to the levels indicated on the x-axis when there are no entrepreneurs. In panel B, the solid line shows the CEVs (in percent) when there is no highest labor ability ϵ_6 . In both panels, the dotted line shows the CEVs (in percent) in the baseline experiment. Vertical lines indicate optimal levels of associated TMTRs. All results obtained for steady state comparisons.

Optimal Tax Rates under Alternative Specifications

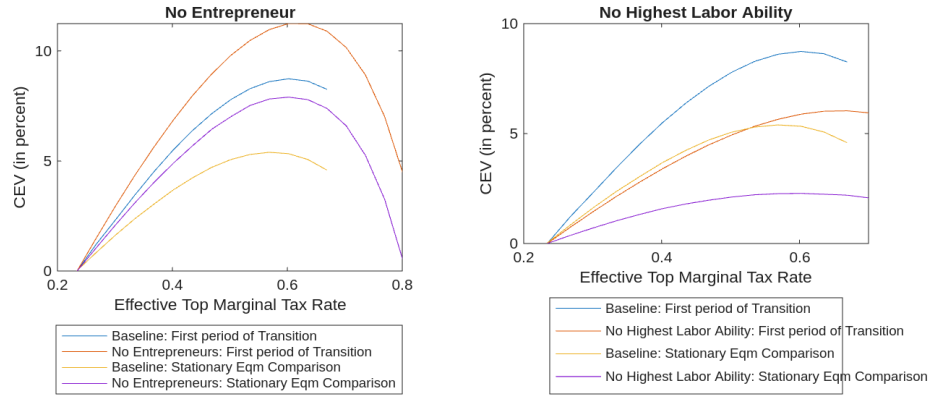


Figure 8: Figure 6 of Bruggemann (2021). The x-axis is extended to an effective top marginal rate of 80.3 percent, further than the original, so that the peaks of the alternative economies are interior rather than at the edge of the plotted range.

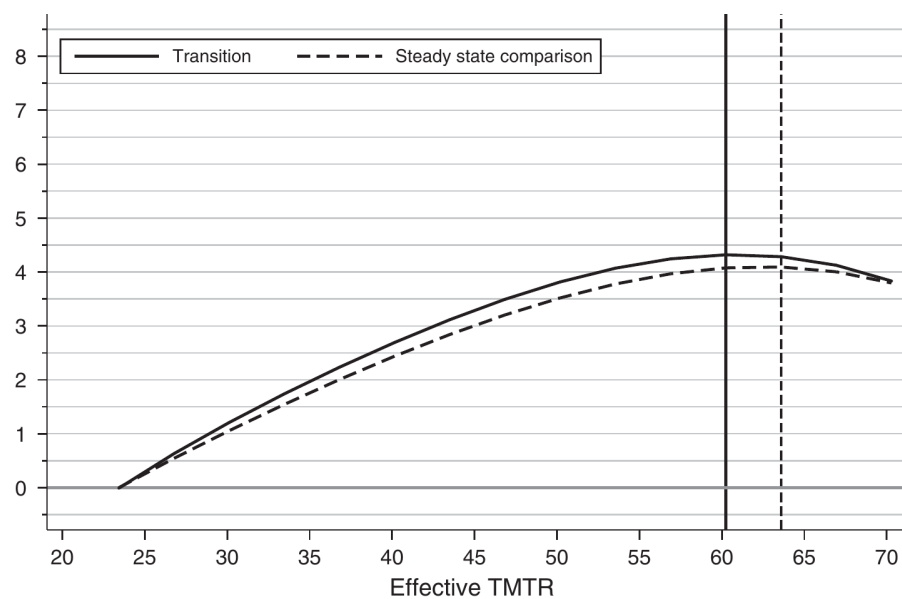


FIGURE A1. REVENUE-MAXIMIZING TOP MARGINAL TAX RATE

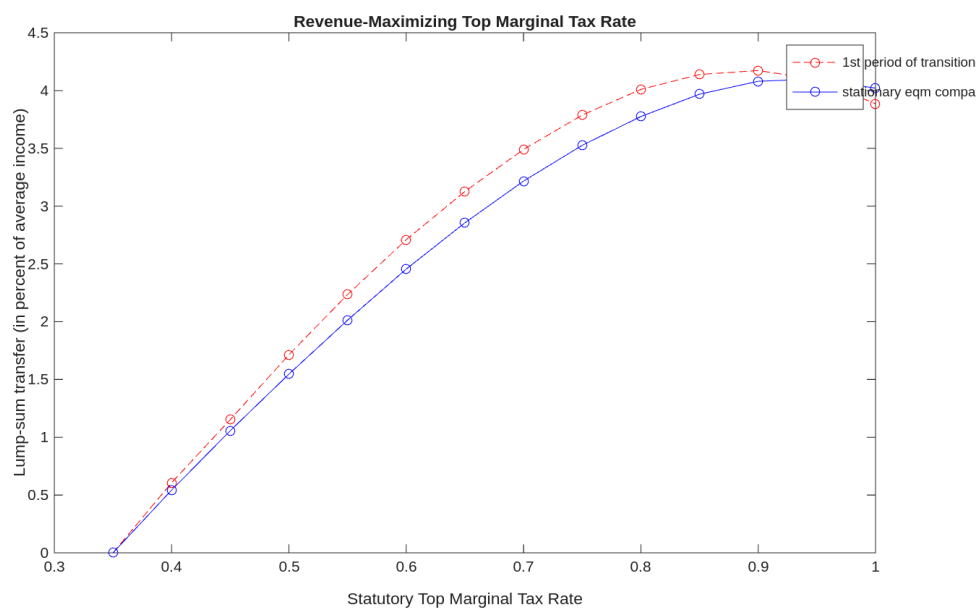


Figure 9: Figure A1 of Bruggemann (2021), the Laffer curve. Revenue is maximized at an effective top marginal rate of 63.6 percent in the steady state and 60.2 percent in the first period of the transition, against 62 and 60 percent in the original.

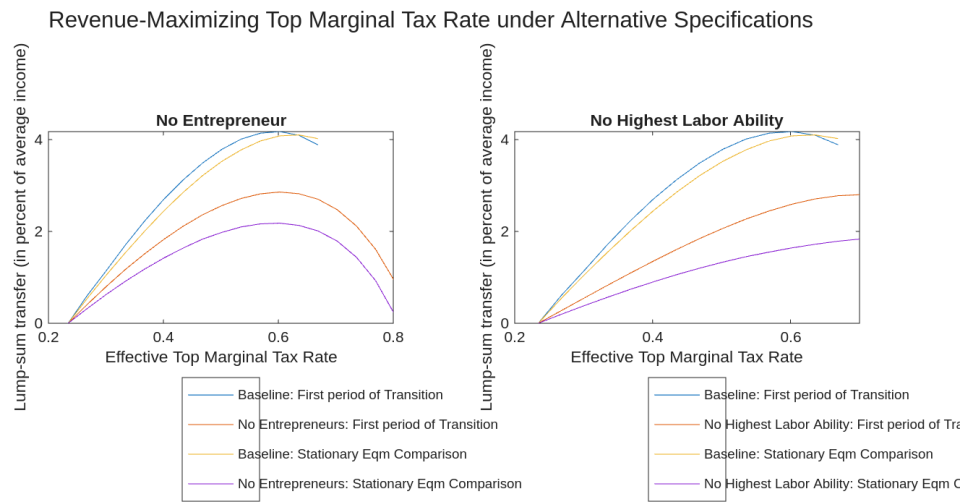


Figure 10: Not a figure from Bruggemann (2021). It shows the Laffer curves for the no-entrepreneur models, so it is to Figure 8 what Figure 9 is to Figure 1.

TABLE 3—TARGETS: DATA AND MODEL

	Target	Model
Overall economy		
Capital-output ratio	2.65	2.65
Top 1 percent income share	0.17	0.17
Ratio of median net worth E/W	7.26	7.69
Wealth Gini	0.85	0.84
Average federal income tax rate	0.12	0.12
Entrepreneurs		
Fraction of entrepreneurs	0.07	0.07
Entry rate	0.02	0.02
Exit rate	0.22	0.22
Entrepreneurs' share of total income	0.17	0.22
Entrepreneurs' income Gini	0.65	0.62
Share of entrepreneurs among top 1 percent income	0.35	0.35
Share of hiring entrepreneurs	0.66	0.64
Workers		
Average working time	1.00	1.00
Workers' income Gini	0.51	0.52

Calibration targets and results

	Target	Model
Overall Economy		
Capital-output ratio	2.65	2.67
Top 1 percent income share	0.17	0.17
Ratio of median net worth E/W	7.26	5.66
Wealth Gini	0.85	0.84
Average Federal Income Tax rate	0.12	0.16
Entrepreneurs		
Fraction of entrepreneurs	0.07	0.07
Entry rate	0.02	0.02
Exit rate	0.22	0.24
Entrepreneurs share of total income	0.17	0.22
Entrepreneurs income Gini	0.65	0.62
Share of Entrepreneurs among top 1 percent income	0.35	0.35
Share of entrepreneurs who hire	0.66	0.64
Workers		
Average working time	1.00	1.00
Workers income Gini	0.51	0.52

Note: Targets not checked as part of replication. Renamed "Share of hiring entrepreneurs" to "Share of entrepreneurs who hire", as I find this clearer. We include the revenue from

$\tau_{a,s}$

in the federal income tax revenue rates, since it is a federal tax and levied on income, original appears not to include it. "Ratio of median net worth E/W" is the median assets of entrepreneurs (young households who choose to be entrepreneurs) divided by the median assets of everyone else, namely young workers together with retirees. Grouping the retirees in with the workers follows B2021, whose codes build this denominator as the non-entrepreneur share of the asset distribution (benchmark.f90, subroutine medw_ratio), rather than as young workers alone. The E group is subtle. Our entrepreneurs are the young households who *choose* to be entrepreneurs this period, because in this replication the occupation is a decision rather than an endogenous state. B2021 instead has the occupation

TABLE 4—DISTRIBUTIONS OF INCOME AND WEALTH

	Income				Wealth			
	Gini	Top 1%	Top 5%	Top 10%	Gini	Top 1%	Top 5%	Top 10%
All households								
Data	0.549	17.2	33.7	44.4	0.846	34.1	60.9	74.4
Model	0.572	17.0 ^a	42.2	49.8	0.842 ^a	21.6	60.9	77.7
Entrepreneurs								
Data	0.650	21.1	42.2	55.1	0.771	25.5	50.1	64.1
Model	0.621 ^a	9.7	32.4	49.1	0.709	14.6	39.9	55.3
Workers and retirees								
Data	0.518	14.5	29.7	40.6	0.832	31.4	57.3	71.3
Model	0.519 ^a	17.4	35.5	44.2	0.850	22.6	64.2	78.9

^aTargeted.

Source: Data on income and wealth distributions are obtained from the Survey of Consumer Finances 2010 (Board of Governors of the Federal Reserve System 2010a, b).

Distributions of Income and Wealth

	Income				Wealth			
	Gini	Top 1%	Top 5%	Top 10%	Gini	Top 1%	Top 5%	Top 10%
All Households								
Data	0.549	17.2	33.7	44.4	0.846	34.1	60.9	74.4
Model	0.572	17.0	42.3	50.0	0.843	21.4	60.9	77.8
Entrepreneurs								
Data	0.650	21.1	42.2	55.1	0.771	25.5	50.1	64.1
Model	0.621	9.7	32.5	49.3	0.719	15.4	40.5	56.3
Workers and Retirees								
Data	0.518	14.5	29.7	40.6	0.832	31.4	57.3	71.3
Model	0.519	17.4	35.6	44.5	0.851	22.3	64.0	79.2

Note: Data not checked during replication

Figure 12: Table 4 of Bruggemann (2021)

TABLE 5—AVERAGE TAX RATES FOR TOP PERCENTILES OF THE INCOME DISTRIBUTION

	Top 1%	Top 3%	Top 5%	Top 10%
Data	23.4	21.9	20.6	18.5
Model	21.6	20.9	20.0	18.5

Note: Average tax rates are computed by dividing total income tax by taxable income.

Source: Data are obtained from the Internal Revenue Service (Statistics of Income SOI Tax Stats 2001–2017).

Average Tax Rates for Top Percentiles of the Income Distribution (%)				
	Top 1%	Top 3%	Top 5%	Top 10%
Data	23.4	21.9	20.6	18.5
Model	26.3	25.6	24.7	23.0

Note: Data not checked during replication. We include the revenue from

$$\tau_{u_s}$$

in the average tax rates, since it is a federal tax and levied on income, original appears not to include it.

Figure 13: Table 5 of Bruggemann (2021)

TABLE 6—FIRM SIZE DISTRIBUTION: DATA AND MODEL

	Data	Model
Share of hiring entrepreneurs	0.661	0.636 ^a
1–5 employees	0.692	0.588
6–10 employees	0.119	0.179
11–20 employees	0.065	0.102
More than 20 employees	0.125	0.131

^aTargeted.*Source:* Data on firm size distribution are obtained from the SCF 2010.

Firm Size Distribution: Data and Model

	Data	Model
Share of entrepreneurs who hire	0.661	0.638
1-5 employees	0.692	0.588
6-10 employees	0.119	0.182
11-20 employees	0.065	0.100
More than 20 employees	0.125	0.131

Note: Data not checked during replication. Note: Rows 2-to-5 are expressed as a fraction of row 1 (so rows 2-5 sum to one). The model has no notion of a headcount of employees, only of efficiency units of hired labor n . Following B2021 (Stata/Dofiles/table6.do) we take one efficiency unit to be one employee and bin n as a headcount rounded to the nearest integer, so the cutoffs are 5.5, 10.5 and 20.5 rather than 5, 10 and 20, and n is used as it is rather than rescaled by average labor supply.

Figure 14: Table 6 of Bruggemann (2021)

TABLE 7—AGGREGATE CHANGES BETWEEN STEADY STATES AFTER INCREASING THE EFFECTIVE TMTR TO 60 PERCENT

	Y	K	N	T	r	w
$\tau^{\max} = 60\%$, GE (percent)	-10.7	-27.1	-5.1	19.6	56.6	-7.4
$\tau^{\max} = 60\%$, PE (percent)	-15.4	-38.8	-3.4	15.5	0.0	0.0

Note: PE stands for partial equilibrium (prices at benchmark level), GE for general equilibrium (prices adjust).

Aggregate Changes Between Stationary Equilibria after Increasing the Effective TMTR to 60.2 percent

	Y	K	N	T	r	w
$\tau^{\max} = 60.2\%$, GE (percent)	-10.8	-27.5	-4.5	17.0	58.8	-7.5
$\tau^{\max} = 60.2\%$, PE (percent)	-16.0	-39.5	-3.6	13.4	0.0	0.0

Note: PE stands for partial equilibrium (prices at benchmark level), GE for general equilibrium (prices adjust).

Figure 15: Table 7 of Bruggemann (2021)

TABLE 8—AGGREGATE CHANGES BETWEEN STEADY STATES AFTER INCREASING THE EFFECTIVE TMTR TO 60 PERCENT: BY SECTOR

	<i>Y</i>	<i>K</i>	<i>N</i>	# Ent.
Corporate sector				
$\tau^{\max} = 60\%$, GE (percent)	−19.5	−31.1	−13.0	
$\tau^{\max} = 60\%$, PE (percent)	−12.5	−43.3	+8.4	
Entrepreneurial sector				
$\tau^{\max} = 60\%$, GE (percent)	−0.9	−13.8	+7.1	+4.1
$\tau^{\max} = 60\%$, PE (percent)	−18.7	−23.9	−21.5	+1.5

Note: PE stands for partial equilibrium (prices at benchmark level), GE for general equilibrium (prices adjust).

Aggregate Changes Between Stationary Equilibria after Increasing the Effective TMTR to 60.2 percent: By S

	<i>Y</i>	<i>K</i>	<i>N</i>	#E
Corporate Sector				
$\tau^{\max}=60.2\%$, GE (percent)	-20.1	-31.8	-13.6	
$\tau^{\max}=60.2\%$, PE (percent)	-13.1	-44.0	7.9	
Entrepreneurial Sector				
$\tau^{\max}=60.2\%$, GE (percent)	-0.4	-13.2	7.9	4.3
$\tau^{\max}=60.2\%$, PE (percent)	-19.2	-24.7	-22.0	1.3

Note: PE stands for partial equilibrium (prices at benchmark level), GE for general equilibrium (prices adjust).

Figure 16: Table 8 of Bruggemann (2021)

TABLE 9—RELATIVE FACTOR ALLOCATION IN CORPORATE AND ENTREPRENEURIAL SECTOR
IN THE BENCHMARK STEADY STATE AND AFTER INCREASING EFFECTIVE TMTR
TO 60 PERCENT

	Capital		Labor	
	Corporate	Entrepreneurial	Corporate	Entrepreneurial
Benchmark (percent)	76.7	23.3	60.5	39.5
$\tau^{\max} = 60\%$, GE (percent)	72.5	27.5	55.4	44.6

Relative Factor Allocation in Corporate and Entrepreneurial Sector in the Benchmark Stationary
Equilibrium and After Increasing the Effective TMTR to 60.2 percent

	Capital		Labor	
	Corporate	Entrepreneurial	Corporate	Entrepreneurial
Benchmark (percent)	76.9	23.1	60.7	39.3
$\tau^{\max}=60.2\%$, GE (percent)	72.4	27.6	55.3	44.7

Note: GE means general equilibrium (prices adjust); only the general equilibrium is reported here. The labor shares are shares of *market* labor, that is, of total efficiency units of labor hired by the two sectors; entrepreneurs own hours, which are supplied inelastically at $\bar{l}$ and are not hired on the labor market, are excluded from the denominator. This follows the original Fortran codes, in which corporate labor is `totlcorp=toteffl-hiredlabe` and entrepreneurial labor is `hiredlabe`.

Figure 17: Table 9 of Bruggemann (2021)

TABLE 10—EFFECTS ON ENTREPRENEURIAL SECTOR

	Entry (in percent)	Exit (in percent)	Fraction (in percent)
Benchmark	2.3	22.2	7.2
$\tau^{\max} = 60\%$, steady state (percent)	+3.3	−1.3	+4.1
$\tau^{\max} = 60\%$, transition (percent)	−2.5	+4.3	−1.5

Notes: The second line shows percentage changes in the entry and exit rate to and from entrepreneurship as well as the total fraction of entrepreneurs when comparing steady states. The third line shows changes in the first period of the transition.

Effects on Entrepreneurial Sector			
	Entry (in percent)	Exit (in percent)	Fraction (in percent)
Benchmark	1.8	23.9	7.2
$\tau^{\max}=60.2\%$, stationary eqm	3.2	-1.4	4.3
$\tau^{\max}=60.2\%$, transition path	0.3	-0.9	-2.0

Note: The benchmark row reports levels, the other two rows report percent changes relative to the benchmark. The stationary eqm row compares the two stationary equilibria; the transition path row uses the first period of the transition.

Figure 18: Table 10 of Bruggemann (2021)